



Report on

RELIABILITY PREDICTION

OF

Camera Model No.: HC-IPC-DI3212DVZK-N5-O
(CCTV CAMERA)

as per Telcordia SR-332, Issue 4



CENTRE FOR RELIABILITY,

STQC Directorate
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2. The analysis report provided is solely based on a theoretical estimation for predicting design reliability.
3. The accuracy and validity of the design reliability prediction cannot be guaranteed due to its theoretical nature.
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1.0 SCOPE

- 1.1** Service request no. : 2602330
- 1.2** Name & Address of the Client : Hi-Focus Electronics India Pvt Ltd
3, Sunguvar Agraharam St,
Chindatripet, Chintadripet, Chennai,
Tamil Nadu 600002.
- 1.3** Description : **CCTV CAMERA**
Make : Hi-Focus Electronics India Pvt Ltd
Model No. : HC-IPC-DI3212DVZK-N5-O
Serial No. : HFP25100543120377
- 1.4** Project : Received Date : 11-05-2026
Completed Date : 12-06-2026
- 1.5** (a) Standard/ Analysis / Test Procedure Ref. : Telcordia SR-332, Issue 4
(b) Scope : Reliability Prediction by Parts Count Method

V. Jambulingam
Approved by 12/6/26

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2.0 INTRODUCTION

This report presents the results of Reliability Prediction Analysis performed on **Camera Model No.: HC-IPC-DI3212DVZK-N5-O** as per Telcordia SR-332, Issue 4 as depicted in Figure 1.0, 2.0, 3.0, 4.0, 5.0, and 6.0. This exercise was performed under **Ground, Fixed, Controlled (GB)** environment as requested by **Hi-Focus Electronics India Pvt Ltd.**

Design Reliability Prediction is the process of quantitatively assessing whether a proposed or actual equipment design will meet a specified reliability requirement. Design Reliability Prediction has various applications in the design and development process including substantiating quantitative design comparisons with competitive products, selecting warranty periods, advertising and marketing. It provides guidance, relative to expected inherent reliability of a given design.

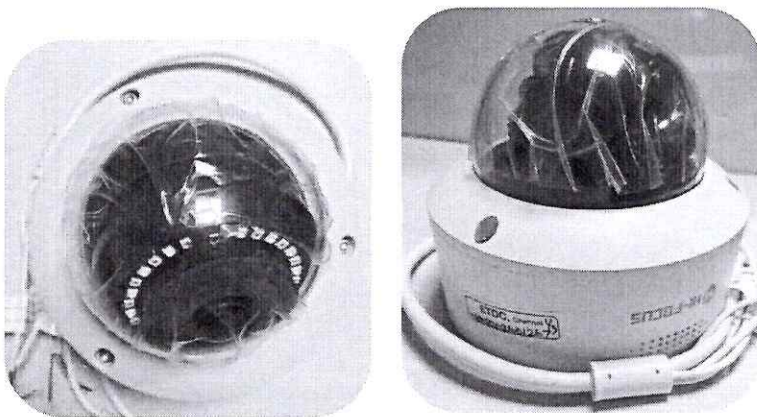


Figure 1.0 (a & b) Front & Top view of camera



Figure 2.0 (a) Markings on camera

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Camera Model No.: HC-IPC-DI3212DVZK-N5-O

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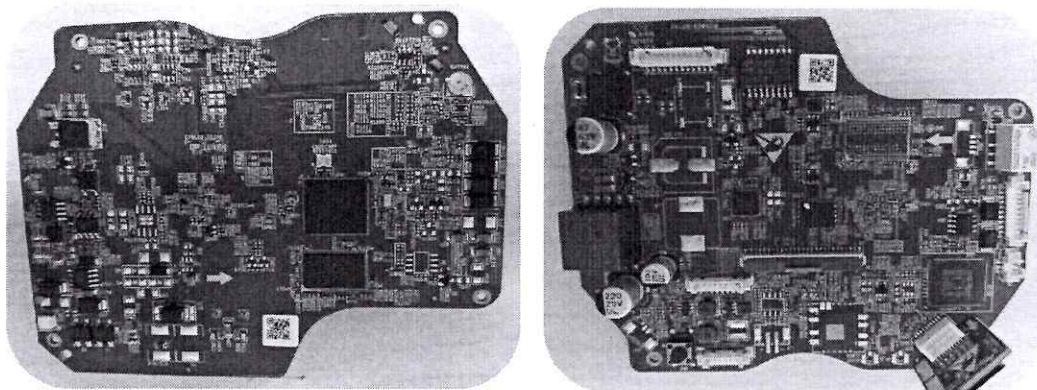


Figure 3.0 (a & b) Front & Rear View of PCB 1

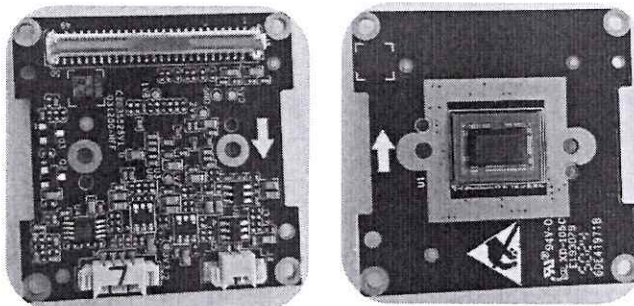


Figure 4.0 (a & b) Front & Rear View of PCB 2

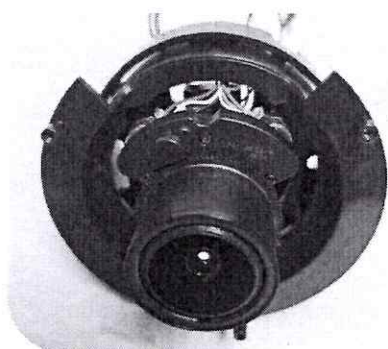


Figure 5.0 (a) Motorized Lens

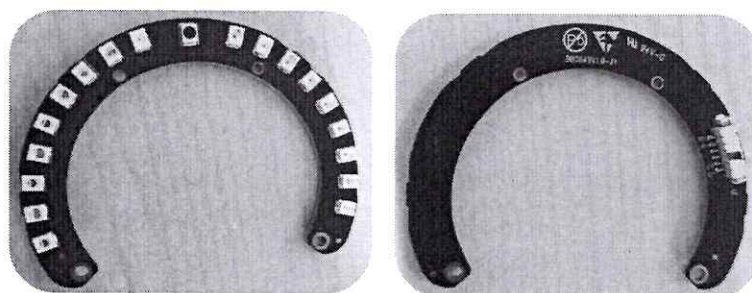


Figure 6.0 (a & b) IR Board & LEDs

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3.0 APPLICABLE DOCUMENTS AND TOOLS

Telcordia SR-332, Issue 4 - Reliability Prediction Procedure for Electronic Equipment

4.0 RELIABILITY PREDICTION OBJECTIVES

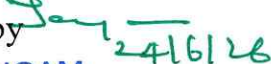
Objectives of this analysis are to estimate the Mean Time between Failure (MTBF) of **Camera Model No.: HC-IPC-DI3212DVZK-N5-O** using Telcordia SR-332, Issue 4 and to obtain quantitative design information as applicable. The main objectives are:

- a. Provide an estimate of the steady-state failure rates for device design based on generic reliability parameters and obtain feasibility evaluation.
- b. Provide feedback for designers.
- c. Identify potential problem areas during design review.
- d. Provide MTBF of the system and sub-assemblies and failure rate of system/sub-assemblies/ Components.

5.0 SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method - ASSUMPTIONS

Reliability Prediction using SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method for reliability prediction operates under the following assumptions: assumes a serial relationship between parts of the system, in which, failure of any part constitutes system failure. The model also assumes that

- a. At the device level, it is assumed that no detailed reliability data is available. Predictions are based on generic reliability parameters.
- b. Predictions are made using generic reliability parameters as specified in the relevant sections of SR-332, which are applied to estimate steady state failure rates.
- c. At the unit level, failure rate predictions are based solely on the failure rates of the individual devices that compose the unit, without considering interactions between devices.
- d. This method primarily estimates steady-state failure rates and does not specifically account for infant mortality failures (early-life failures) or wear out failures (end-of-life failures). It assumes a constant failure rate.

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6.0 SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method: Factors

Method 1: Unit Steady-State Failure Rate Using the Parts Count Method primarily estimates reliability by calculating the steady-state failure rate of a device using a set of generic parameters. It takes into account the following factors:

- 6.0.1 Number of component of each Type
- 6.0.2 Mean Generic Steady-State Failure Rate
- 6.0.3 Temperature factor
- 6.0.4 Environmental factor
- 6.0.5 Quality factor
- 6.0.6 Electrical Stress Factor

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7.0 ENVIRONMENTAL CONDITIONS

7.1 STORAGE AND OPERATIONAL ENVIRONMENT

The Reliability Prediction has been performed using **Ground, Fixed, Controlled (GB)** Environment based on the operational conditions of **Camera Model No.: HC-IPC-DI3212DVZK-N5-O**.

GROUND, FIXED, CONTROLLED (GB):

The environment represents a stationary, land-based installation located in a protected or climate-controlled facility. It is characterized by low vibration and shock stresses, low atmospheric variations, and low temperature-cycling stresses, with the understanding that a building or ruggedized enclosure provides consistent protection from weather and contaminants.

TEMPERATURE

Reliability Prediction has been performed on **Camera Model No.: HC-IPC-DI3212DVZK-N5-O** at the **Ambient Temperature of 40 °C for Ground, Fixed, Controlled (GB) Environment**.

7.2 QUALITY LEVEL : Level- 1

The unit has been evaluated under the commercial-grade component Quality level-1 as mentioned by the client in bill of material. The components used in the unit were procured without undergoing thorough device qualification or lot-to-lot controls by the equipment manufacturer.


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8.0 REPORTS

Report includes the following information-

- a) Components/ Parts descriptions
- b) Failure Rate
- c) Pi (π) Factors
- d) Parameters details

Refer Annexure- I.

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9.0 System Analysis Summary

The overall estimated MTBF and Failure Rate as per Telcordia SR-332 of **Camera Model No.: HC-IPC-DI3212DVZK-N5-O**, submitted for reliability prediction analysis by **Hi-Focus Electronics India Pvt Ltd** are summarized as follows:

SYSTEM: Camera Model No.: **HC-IPC-DI3212DVZK-N5-O**

Environment	Ambient Temperature	Failure Rate (Failures/Million Hours)	MTBF (Hrs)
Ground, Fixed, Controlled (GB)	40°C	4.172301	239675.91

- Design reliability prediction analysis was carried out based on the failure rates of components used in the camera (refer Annexure I), following a physical verification of components and in accordance with Telcordia SR-332, Issue 4.
- The camera contains three (03) nos of synchronous/stepper/shutter motors. However, failure rate data for these motors is not available in the Telcordia standard or in the reliability prediction software used. Therefore, for the purpose of the design reliability prediction, the failure rate for each motor has been conservatively assumed as one (1) FIT.
- MTBF (Mean Time Between Failures) calculations have been performed using the component details—such as type, quantity, electrical and environmental parameters, stress levels, and quality grades—as provided by the client in their submitted Bill of Materials.

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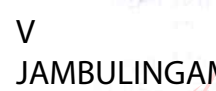
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10.0 Abbreviation Used:

- a) MTBF : Mean Time Between Failure
- b) λ_G : Base failure rate for the generic part (Ground fixed base rate)
- c) σ_G : Standard deviation or uncertainty in base failure rate (generic)
- d) π_Q : Quality factor (related to part/process quality level)
- e) π_E : Environmental factor (based on the operating environment, e.g., ground benign, airborne)
- f) π_T : Temperature factor (influenced by ambient/operating temperature)
- g) π_S : Stress factor (e.g., voltage or power stress ratio)
- h) λ_{SS} : Subsystem or system-level failure rate (or stress-adjusted failure rate)
- i) σ_{SS} : Standard deviation or uncertainty in system/subsystem failure rate

- End of Report -


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Annexure - I



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		Reliability Prediction Report				Calculation Model		Telcordia Issue 4	
Name		CCTV CAMERA				Environment		Ground, Fixed, Controlled (GB)	
Description		Make : HI-FOCUS				Ambient Temperature, Celsius		40.00	
		Model Number: HC-IPC-DI3212DVZK-N5-O				Operating Profile		Telecommunications	
		Serial Number : HFP25100543120377				Failure Rate Units		Failures/Million Hours	
Quantity		1				Failure Rate		4.172301	
Reference Designator						MTBF Units		Hours	
						MTBF		239675.91	

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.1	R0402RXX330XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R7,R8,R10,R 11,R12,R13	330±1%-0402-(- 55°C~155°C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.2	R0402RXX472XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5,R15,R28,R36, R65,R72,R90,R9 1	4.7K±1%-0402-(- 55°C~155°C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.3	R0402RXX330XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R20,R21,R42, R43,R57,R58	330±1%-0402-(- 55°C~155°C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)



Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.4	R0402RXX202XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R74, R94	2K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π F: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.5	R0402RXX223XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R24	22K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π F: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.6	R0402RXX105XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R70, R95	1K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π F: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.7	R0402RXX105XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R27	1M $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Mohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π F: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.8	R0402RXX103XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R7,R11,R29,R30, R39,R93	10K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.9	R0402RXX104XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R31,R89,R96	100K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.10	R0402RXX241XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R38	2400 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.11	R0402JXX0000P 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R32,R34,R44	00 $\pm$ 5%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	3	0.000152	4.562529E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.12	R0402RX183XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R33,R37,R69,R88	18K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.13	R0402RX152XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R92	1.5K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1.5, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.14	R0402RX332XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R18	3.3K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.15	R0402RX473XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R87	47K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.16	R0402RXX751XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R35	7500±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, nS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.17	R0402RXX153XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R73	15K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, nS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.18	R0402RXX471XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R706, C1401	4700±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, nS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.19	R0402RXX474XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R41, R1419	470K±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, nS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.20	R0402RXX472XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R45,R52,R63,R8 6,R87,R90,R93,R 124,R150,R191, R220,R227,R274 R277,R703,R71 2,R737,R742,R7 46,R803,R809,R 812,R815,R818, R822,R823,R824 R859,R860,R86 3,R897,R1301,R 1402	4,7K $\pm$ 1%-0402- (-55°C~155°C)	33	0.000163	0.005374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.21	R0402RXX330XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5,R6,R10,R11, R17,R18,R25,R6 1,R94,R101,R15 3,R157,R253,R5 17,R518,R804,R 811,R825,R857, R858,R861,R864 R865,R866,R86 7,R868	330 $\pm$ 1%-0402- (-55°C~155°C)	26	0.000164	0.004263	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.22	R0402RXX202XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R20,R91,R99,R1 40,R181,R183,R 186,R188,R246, R896,R1403	2K $\pm$ 1%-0402- (-55°C~155°C)	11	0.000169	0.001861	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.23	R0402RXX223XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R141,R724	22K $\pm$ 1%-0402- (-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)



Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.24	R0402RXX2222XF 161TA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R78,R603	2.2K±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.25	R0402RXX102XF 161TA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R21,R23,R27,R1 33,R134,R138,R 147,R149,R151, R171,R217,R248 ,R345,R734,R74 4,R748,R830,R1 003,R1005	1K±1%-0402-(-55°C~155°C)	19	0.000166	0.003146	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.26	R0402RXX103XF 161TA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R29,R50,R10 0,R130,R143,R1 45,R219,R236,R 238,R239,R247, R601,R602,R709 ,R722,R736,R73 8,R740,R741,R8 31,R1013,R1015 ,R1016,R1201,R 2621	10K±1%-0402-(-55°C~155°C)	26	0.000164	0.004263	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.27	R0402RXX104XF 161TA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R721,R801	100K±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, λG: 0.08, σG: 0.02, λSS: 0.15425991, σSS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.28	R0402RXX241XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1007,R1008	2400±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	ΛG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.64274964, ΛG: 0.08, σG: 0.02, ΛSS: 0.15425991, σSS: 0.038564978, Temperature: 40
1.29	CR-05JL7--10R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R113,R114,R115,R116,R120,R262,R899	100±5%-0805-(-55°C~155°C)	7	0.000156	0.001091	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	ΛG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, ΛG: 0.08, σG: 0.02, ΛSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.30	CR-05FL7--10K	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R117,R118	10K±1%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: kohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	ΛG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, ΛG: 0.08, σG: 0.02, ΛSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.31	CR-05FL7--100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R178	1000±1%-0805-(-55°C~155°C)	1	0.000185	1.850080E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	ΛG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.57926223, ΛG: 0.08, σG: 0.02, ΛSS: 0.13902294, σSS: 0.034755734, Temperature: 40
1.32	CR-05JL7--0R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R35,R107,R112	00±5%-0805-(-55°C~155°C)	4	0.000147	5.890311E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	ΛG: 0.08, σG: 0.02, πQ: 3, πE: 1, πT: 1, πS: 0.52750338, ΛG: 0.08, σG: 0.02, ΛSS: 0.12660081, σSS: 0.031650203, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.33	R0402JXX0000P 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R2,R13,R16,R76, R160,R166,R240 ,R637,R646,R64 7,R650,R651,R6 65,R723,R735,R 739,R750,R756, R766,R819,R826 ,R827,R833,R10 01,R1002,R1203 ,R1423,R1599,R 1705,R1723	00 $\pm$ 5%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	30	0.000135	0.004062	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40
1.34	R0402RXX100XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R121,R229,R230 ,R231,R233,R23 4,R235,R2622	100 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.35	R0402RXX101XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R14,R19,R24,R2 8,R814,R817,R1 404	1000 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.36	CR-03FL7---75R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R290,R291,R421 ,R422	750 $\pm$ 1%-0603-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	4	0.000535	0.002139	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 75, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 1.9155408, λ G: 0.08, σ G: 0.02, λ SS: 0.4597298, σ SS: 0.11493245, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.37	R0402RXX183XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R84, R729, R1110	18K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.38	R0402RXX333XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R743, R745	33K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.39	R0402RXX152XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R237	1.5K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1.5, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.40	R0402RXX333XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R142, R170, R224, R717, R832, R898	3.3K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.41	R0402RXX224XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R128	220K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 220, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.42	R0402RXX201XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R139	200K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 200, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.43	R0402RXX153XF 16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R155, R212, R218, R704, R720, R725, R732, R747	15K $\pm$ 1%-0402-(-55°C~155°C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.44	R0402RWX2R2X F16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R273, R286, R287, R289	2.2K $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.45	RL1206FR-070R51L	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R123, R127, R131, R132	510m Ω $\pm$ 1%-1206-(-55°C~155°C)	4	0.000154	6.140516E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.51, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.25, Percentage: 4	Applied Power: 0.01, Rated Power: 0.25, Percentage: 4	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.54991036, λ G: 0.08, σ G: 0.02, λ SS: 0.13197849, σ SS: 0.032994621, Temperature: 40

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Approved By: V. Jambulingam (Head Reliability)



Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.46	RS-05FL7--0R2	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R81	200mR $\pm$ 1%-0805 (-55°C ~155°C)	1	0.000280	2.804510E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.2, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.87809543, λ G: 0.08, σ G: 0.02, λ SS: 0.2107429, σ SS: 0.052685726, Temperature: 40
1.47	CR0402FF56R0G	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1014, R1017	56R $\pm$ 1%-0402 (-55°C ~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 56, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.48	CR0402FF680G	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1018, R1019, R1020, R1021	680R $\pm$ 1%-0402 (-55°C ~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 680, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.49	CC0402KXSR7 BB106	Capacitor	Ceramic	General	C1	10 μ F $\pm$ 10%-16V-X5R-0805	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.50	CC0402KRX7R7 BB104	Capacitor	Ceramic	General	C2	100nF $\pm$ 10%-16V-X7R-0402	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.51	CC0402JRNPO9 BN100	Capacitor	Ceramic	General	C81	10 μ F $\pm$ 50V-NP0-0402	1	0.000055	5.526646E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.52	CC0402KRX7R9 BB102	Capacitor	Ceramic	General	C47	1nF $\pm$ 10%-50V- X7R-0402	1	0.000048	4.815829E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, Δ G: 0.1, σ G: 0.01, Δ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.53	0402X105K6R3C T	Capacitor	Ceramic	General	C30,C31,C32,C5 8	1uF $\pm$ 10%-6.3V- X5R-0402	4	0.000090	3.589336E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.28109921, Δ G: 0.1, σ G: 0.01, Δ SS: 0.084329762, σ SS: 0.0084329762, Temperature: 40
1.54	CC0402KRX7R7 BB104	Capacitor	Ceramic	General	C14,C33,C34,C3 8,C54,C56,C59,C 60,C61,C62,C63, C64,C70,C80,C8 8,C89,C93,C99,C 457	100nF $\pm$ 10%- 16V-X7R-0402	19	0.000093	0.001760	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.2987937, Δ G: 0.1, σ G: 0.01, Δ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.55	0402B103K500C T	Capacitor	Ceramic	General	C11,C13,C20,C4 4	10nF $\pm$ 10%-50V- X7R-0402	4	0.000045	1.813783E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, Δ G: 0.1, σ G: 0.01, Δ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.56	0402X225M6R3 CT	Capacitor	Ceramic	General	C12	2.2uF $\pm$ 20%- 6.3V-X5R-0402	1	0.000288	2.876279E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, Δ G: 0.1, σ G: 0.01, Δ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.57	CL05A475M05 NRNC	Capacitor	Ceramic	General	C3,C6,C7,C9,C1 5,C17,C19,C21,C 23,C25,C27,C36	4.7uF $\pm$ 20%- 6.3V-X5R-0402	12	0.000264	0.003173	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, Δ G: 0.1, σ G: 0.01, Δ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.58	0402X224K100 NT	Capacitor	Ceramic	General	C86,C104	220nF±20%- 10V-X5R-0402	2	0.000163	3.259277E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.4980773, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.14942319$, $\sigma SS: 0.014942319$, Temperature: 40
1.59	CL10A106KP8N NNC	Capacitor	Ceramic	General	C39,C57,C71,C4 36	10uF±10%-10V- X5R-0603	4	0.000319	0.001277	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ $\lambda G: 0.1, \sigma G: 0.01, \lambda SS: 0.3$, $\sigma SS: 0.03$, Temperature: 40
1.60	CL10A226MQ8 NRNC	Capacitor	Ceramic	General	C37,C40,C42,C5 1,C55,C79,C90,C 92,C94,C96,C10 3,C105,C107,C2 32,C233	22uF±20%-6.3V- X5R-0603	15	0.000342	0.005126	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 1.1025427, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.33076281$, $\sigma SS: 0.033076281$, Temperature: 40
1.61	CC0402JRNPO9 BN100	Capacitor	Ceramic	General	C825	10pF±5%-50V- NPO-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808$, $\sigma SS: 0.0050621808$, Temperature: 40
1.62	CC0402JRNPO9 BN180	Capacitor	Ceramic	General	C815,C822,C824	18pF±5%-50V- NPO-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808$, $\sigma SS: 0.0050621808$, Temperature: 40
1.63	CC0402JRNPO9 BN101	Capacitor	Ceramic	General	C181,C991	100pF±5%-50V- NPO-0402	2	0.000063	1.269350E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.19398004, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.058194013$, $\sigma SS: 0.0058194013$, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.64	CC0402KRX7R9 BB102	Capacitor	Ceramic	General	C182,C1408,C1417	1nF±10%-50V-X7R-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16873936, AG: 0.1, σG: 0.01, ASS: 0.050621808, σSS: 0.0050621808, Temperature: 40
1.65	0402B332K500C T	Capacitor	Ceramic	General	C73	3.3nF±10%-50V-X7R-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16873936, AG: 0.1, σG: 0.01, ASS: 0.050621808, σSS: 0.0050621808, Temperature: 40
1.66	CC1206KKX7RD BB102	Capacitor	Ceramic	General	C30,C89,C126,C158,C167,C168,C284,C327	1nF±10%-2KV-X7R-1206	8	0.000041	3.262902E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.13006122, AG: 0.1, σG: 0.01, ASS: 0.039018367, σSS: 0.0039018367, Temperature: 40
1.67	0402N221J500C T	Capacitor	Ceramic	General	C171,C173,C705,C716,C731	220pF±5%-50V-NP0-0402	5	0.000054	2.676154E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.16873936, AG: 0.1, σG: 0.01, ASS: 0.050621808, σSS: 0.0050621808, Temperature: 40
1.68	1206B104K201 NT	Capacitor	Ceramic	General	C159,C166	100nF±10%-200V/250V-X7R-1206	2	0.000113	2.253547E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.3443833, AG: 0.1, σG: 0.01, ASS: 0.10331499, σSS: 0.010331499, Temperature: 40
1.69	1206B105K101C T	Capacitor	Ceramic	General	C161,C163	1uF±10%-100V-X7R-1206	2	0.000301	6.028542E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.92127196, AG: 0.1, σG: 0.01, ASS: 0.27638159, σSS: 0.027638159, Temperature: 40
1.70	CC0805KKX5R7 BB106	Capacitor	Ceramic	General	C58,C62	10uF±10%-16V-X5R-0805	2	0.000912	0.001824	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 16, Percentage: 75	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 16, Percentage: 75	Level I	πQ: 3, πE: 1, πT: 1, πS: 2.7870955, AG: 0.1, σG: 0.01, ASS: 0.83612864, σSS: 0.083612864, Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.71	VZ4221M1ETR-0810	Capacitor	Electrolytic	Aluminum	CE4	220uF±20%-25V-105°C-08*10mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 220, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.95313379, λG : 0.73, σG : 0.17, λSS : 2.087363, σSS : 0.48609823, Temperature: 40
1.72	0603N821J500C-T	Capacitor	Ceramic	General	C282, C283, C325, C326	820pF±5%-50V-NPO-0603	4	0.000054	2.154621E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.16873936, λG : 0.1, σG : 0.01, λSS : 0.050621808, σSS : 0.0050621808, Temperature: 40
1.73	0402X105K683C-T	Capacitor	Ceramic	General	C208, C739, C801, C821	1uF±10%-6.3V-X5R-0402	4	0.000352	0.001408	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	πQ : 3, πE : 1, πT : 1, πS : 1.1025427, λG : 0.1, σG : 0.01, λSS : 0.33076281, σSS : 0.033076281, Temperature: 40

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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)



SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.74	CC0402KRX7R7 BB104	Capacitor	Ceramic	General	C28,C47,C50,C5 3,C57,C68,C88,C 91,C145,C172,C 175,C176,C180, C207,C711,C715 3,C738,C741,C7 47,C751,C753,C 802,C804,C808, C810,C813,C814 C817,C818,C81 9,C820,C823,C8 27,C829,C830,C 831,C832,C835, C836,C838,C850 C852,C853,C85 4,C855,C858,C8 59,C860,C983,C 984,C992,C995, C1002,C1003,C1 004,C1006,C101 2,C1013,C1014, C1015,C1016,C1 017,C1018,C101 9,C1020,C1021, C1022,C1023,C1 024,C1025,C102 6,C1028,C1029, C1030,C1031,C1 032,C1033,C103 4,C1035,C1036, C1037,C1038,C1 039,C1040,C104 1,C1102	100nF $\pm$ 10%- 16V-X7R-0402	89	0.000141	0.012546	Type: General, Quality Level: Level 1, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 16, Percentage: 31.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 16, Percentage: 31.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.4635922, λ G: 0.1, σ G: 0.01, λ SS: 0.13907766, σ SS: 0.013907766, Temperature: 40
1.75	0402B103K500C T	Capacitor	Ceramic	General	C183,C280,C281 ,C601,C1042	10nF $\pm$ 10%-50V- X7R-0402	5	0.000054	2.676154E-004	Type: General, Quality Level: Level 1, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.76	0402X225M6R3 CT	Capacitor	Ceramic	General	CL86,C1461	2.2uF±20%- 6.3V-X5R-0402	2	0.000361	7.214727E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 1.1025427, $\Delta G: 0.1, \sigma G:$ 0.01, $\Delta S: 0.33076281,$ $\sigma SS: 0.033076281,$ Temperature: 40
1.77	0805X226M100 CT	Capacitor	Ceramic	General	C36	22uF±20%-10V- 0805	1	0.000338	3.384465E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ $\Delta G: 0.1, \sigma G: 0.01, \Delta S: 0.3,$ $\sigma SS: 0.03,$ Temperature: 40
1.78	UL1J470M0810 VG	Capacitor	Electrolytic	Aluminum	CE5	47uF±20%-63V- 105°C- Ø8*10mm	1	0.005369	0.005369	Type: Aluminum, Quality Level: Level I, Capacitance: 47, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 1.8749302, $\Delta G: 0.73, \sigma G:$ 0.17, $\Delta S: 4.1060971, \sigma SS:$ 0.9562144, Temperature: 40
1.79	UT1E101M0607 VG	Capacitor	Electrolytic	Aluminum	CE3	100uF±20%- 25V-105°C- Ø6.3*7.7mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 100, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.95313379, $\Delta G: 0.73, \sigma G:$ 0.17, $\Delta S: 2.087363, \sigma SS:$ 0.48609823, Temperature: 40
1.80	CL05A475MQ5 NRNC	Capacitor	Ceramic	General	C209,C805,C807 ,C809,C811,C81 6,C828,C857,C1 402,C1403	4.7uF±20%- 6.3V-X5R-0402	10	0.000344	0.003442	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 1.1025427, $\Delta G: 0.1, \sigma G:$ 0.01, $\Delta S: 0.33076281,$ $\sigma SS: 0.033076281,$ Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.81	CL10A106KP8N N1C	Capacitor	Ceramic	General	C109,C704,C714, C728,C735,C74 6,C812,C826,C8 51,C856,C985,C 989,C993,C1001 ,C1027	10uF±10%-10V- X5R-0603	15	0.000310	0.004649	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 1,$ $\lambda G: 0.1, \sigma G: 0.01, \lambda S: 0.3,$ $\sigma S: 0.03, \text{Temperature:}$ 40
1.82	CL21A106KAYN N1E	Capacitor	Ceramic	General	C63,C92,C156,C 177,C178,C179, C743	10uF±10%-25V- X5R-0805	7	0.000290	0.002028	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.92127196, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda S: 0.27638159,$ $\sigma S: 0.027638159,$ Temperature: 40
1.83	CL10A226MQ8 NRNC	Capacitor	Ceramic	General	C29,C70,C706,C 710,C717,C730, C732,C737,C750 ,C803	22uF±20%-6.3V- X5R-0603	10	0.000344	0.003442	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 1.1025427, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda S: 0.33076281,$ $\sigma S: 0.03076281,$ Temperature: 40
1.84	HCM1206X7R10 3K102PT	Capacitor	Ceramic	General	C169	10nF±10%- 1000V-X7R- 1206	1	0.000044	4.356988E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 0, Rated Voltage: 1000, Percentage: 0	AC Peak Voltage: 0, Applied DC Voltage: 0, Rated Voltage: 1000, Percentage: 0	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.1287349, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda S: 0.038620471,$ $\sigma S: 0.0038620471,$ Temperature: 40
1.85	CL1402-1009- 211	Connection	Multi-Pin		J2	193-F093K230- B06	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	$\lambda G: 0.4, \sigma G: 0.1, \pi Q: 3, \pi E:$ 1, $\pi T: 1, \lambda S: 1.2, \sigma S: 0.3,$ Temperature: 40
1.86	510-628-11	Connection	Multi-Pin		J6	2*10*0.8mm- 5.2mm	1	0.003194	0.003194	Quality Level: Level I, Number of Pins: 20		Level I	$\lambda G: 0.8, \sigma G: 0.2, \pi Q: 3, \pi E:$ 1, $\pi T: 1, \lambda S: 2.4, \sigma S: 0.6,$ Temperature: 40
1.87	11-1832002-330	Connection	Multi-Pin		J4	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	$\lambda G: 0.08, \sigma G: 0.02, \pi Q: 3,$ $\pi E: 1, \pi T: 1, \lambda S: 0.24, \sigma S:$ 0.06, Temperature: 40
1.88	1.25T-1-WT-4A	Connection	Multi-Pin		J2	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	$\lambda G: 0.16, \sigma G: 0.04, \pi Q: 3,$ $\pi E: 1, \pi T: 1, \lambda S: 0.48, \sigma S:$ 0.12, Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.89	60076-5007-1001	Connection	Multi-Pin		J5	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.90	1.25T-1-LT-10A	Connection	Multi-Pin		J6	10*1.25mm	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λG : 0.4, σG : 0.1, πQ : 3, πE : 1, πT : 1, λSS : 1.2, σSS : 0.3, Temperature: 40
1.91	1.25T-1-LT-8A	Connection	Multi-Pin		J11	8*1.25mm	1	0.001278	0.001278	Quality Level: Level I, Number of Pins: 8		Level I	λG : 0.32, σG : 0.08, πQ : 3, πE : 1, πT : 1, λSS : 0.96, σSS : 0.24, Temperature: 40
1.92	1.25T-1-LT-6A	Connection	Multi-Pin		J3	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λG : 0.24, σG : 0.06, πQ : 3, πE : 1, πT : 1, λSS : 0.72, σSS : 0.18, Temperature: 40
1.93	1.25T-1-LT-2A	Connection	Multi-Pin		J8	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λG : 0.08, σG : 0.02, πQ : 3, πE : 1, πT : 1, λSS : 0.24, σSS : 0.06, Temperature: 40
1.94	1.25T-1-LT-12A	Connection	Multi-Pin		J5	12*1.25mm	1	0.001916	0.001916	Quality Level: Level I, Number of Pins: 12		Level I	λG : 0.48, σG : 0.12, πQ : 3, πE : 1, πT : 1, λSS : 1.44, σSS : 0.36, Temperature: 40
1.95	510-627-11	Connection	Multi-Pin		J2	2*10*0.8mm-5.2mm	1	0.003194	0.003194	Quality Level: Level I, Number of Pins: 20		Level I	λG : 0.8, σG : 0.2, πQ : 3, πE : 1, πT : 1, λSS : 2.4, σSS : 0.6, Temperature: 40
1.96	60076-5007-1001	Connection	Multi-Pin		J9	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.97	1.25T-1-LT-6A	Connection	Multi-Pin		J1	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λG : 0.24, σG : 0.06, πQ : 3, πE : 1, πT : 1, λSS : 0.72, σSS : 0.18, Temperature: 40
1.98	NRS3015T2R2M NGH	Inductor	Coil	Chip/Ceramic Filter	L702,L703,L704	2.2uH-1.48A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λSS : 0.3, σSS : 0.21, Temperature: 40
1.99	HXNR4030B-4R7M	Inductor	Coil	Chip/Ceramic Filter	L3,L705	4.7uH-2A	2	0.000499	9.980442E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λSS : 0.3, σSS : 0.21, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.100	0.47mH-1.8A	Inductor	Coil	Chip/Ceramic Filter	L2	0.47mH-1.8A	1	0.000581	5.806969E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λSS : 0.3, σSS : 0.21, Temperature: 40
1.101	RS1M-F1-0002	Semiconductor	Diode	General Purpose, Si	D25	1000V-1A-SMA	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.2, Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.41895155, λG : 0.33, σG : 0.23, λSS : 0.41476203, σSS : 0.28907657, Temperature: 40
1.102	P00805B-LV1	Semiconductor	Diode	General Purpose, Si	D16, D17, D31, D85, D1401, D1403	6.5V-4KV-SMB	6	0.004517	0.027101	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.001, Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.23566226, λG : 4.6, σG : 3.3, λSS : 3.2521391, σSS : 2.3330563, Temperature: 40
1.103	15S355	Semiconductor	Diode	General Purpose, Si	D18, D21, D22, D801	80V-0.1A-SOD323	4	0.000454	0.001818	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.31348618, λG : 0.33, σG : 0.23, λSS : 0.31035132, σSS : 0.21630546, Temperature: 40



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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.104	SCD310H	Semiconductor	Diode	General Purpose, Si	D3,D24	100V-3A-2010/SMAF	2	0.001014	0.002028	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.61672421, \lambda G: 0.33, \sigma G: 0.23, \lambda SS: 0.61055697, \sigma SS: 0.42553971, \text{Temperature: } 40$
1.105	SBR6100CTL-13	Semiconductor	Diode	Voltage Regulator, Zener	D15	100V-6A-DPAK	1	0.017551	0.017551	Type: Voltage Regulator, Zener, Quality Level: Level I, Actual Zener Current or Power: 1, Rated Zener Current or Power: 6, Percentage: 16.6666666667	Rated Zener Current or Power: 6, Percentage: 16.6666666667	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.44932896, \lambda G: 9.3, \sigma G: 2.8, \lambda SS: 12.536278, \sigma SS: 3.7743633, \text{Temperature: } 40$
1.106	BV-SMBJ58C2H	Semiconductor	Diode	TVS Surge Suppressor	D23,D27	58V-10us/700us-85MAX-SMB	2	0.002390	0.004781	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.2, Rated Forward Current: 21.37, Percentage: 0.93589143659	Rated Forward Current: 21.37, Percentage: 0.93589143659	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.24102391, \lambda G: 2.55, \sigma G: 0.81, \lambda SS: 1.8438329, \sigma SS: 0.5856881, \text{Temperature: } 40$
1.107	BV-SMBJ20CAI	Semiconductor	Diode	General Purpose, Si	D7	20V-10/1000uS-800W-30V@8/20us-SMB	1	0.006982	0.006982	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.25837757, \lambda G: 4.6, \sigma G: 3.3, \lambda SS: 3.5656104, \sigma SS: 2.5579379, \text{Temperature: } 40$



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.108	TUS03FB	Semiconductor	Diode	TVS Surge Suppressor	D12,D13	3.3V-8/20uS-350W-Max19V-SOD323	2	0.002327	0.004655	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.001, Rated Forward Current: 8, Percentage: 0.0125	Rated Forward Current: 8, Percentage: 0.0125	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.23465534, Δ G: 2.55, σ G: 0.81, Δ SS: 1.7951133, σ SS: 0.57021246, Temperature: 40
1.109	MB6F	Semiconductor	Diode	General Purpose, Si	Z1,Z2	Glass Passivated Single-Phase Bridge Rectifier	2	0.003924	0.007848	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.4, Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Level I	π Q: 3, π E: 1, π T: 1, π S: 2.3869109, Δ G: 0.33, σ G: 0.23, Δ SS: 2.3630417, σ SS: 1.6469685, Temperature: 40
1.110	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, Δ G: 3.44, σ G: 0.75, Δ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.111	S8550	Semiconductor	Transistor	Field Effect, Silicon	Q3	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, Δ G: 3.44, σ G: 0.75, Δ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.112	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2,Q5,Q9	40V-0.5A-SOT23	3	0.003921	0.011762	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, Δ G: 3.44, σ G: 0.75, Δ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.113	MMBT42LT1G	Semiconductor	Transistor	Field Effect, Silicon	Q16	300V/160V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.333333333	Rated Power: 0.3, Percentage: 3.333333333	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.32627979, λG : 3.44, σG : 0.75, λSS : 3.3672075, σSS : 0.73412954, Temperature: 40
1.114	FDS86242	Semiconductor	Transistor	Field Effect, Silicon	Q12	150V-4.1A-SOP8	1	0.010453	0.010453	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 1, Rated Power: 2.5, Percentage: 40	Rated Power: 2.5, Percentage: 40	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.78662786, λG : 3.44, σG : 0.75, λSS : 8.1179995, σSS : 1.7699127, Temperature: 40
1.115	YJS2305A-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q7	20V-5.4A-SOT23-6L	1	0.007293	0.007293	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.3, Rated Power: 1.2, Percentage: 25	Rated Power: 1.2, Percentage: 25	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.54881164, λG : 3.44, σG : 0.75, λSS : 5.6637361, σSS : 1.2348262, Temperature: 40
1.116	ZM930P06T	Semiconductor	Transistor	Field Effect, Silicon	Q6	60V-3A-SOT23	1	0.008907	0.008907	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.5, Rated Power: 1.5, Percentage: 33.333333333	Rated Power: 1.5, Percentage: 33.333333333	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.67032005, λG : 3.44, σG : 0.75, λSS : 6.9177029, σSS : 1.5082201, Temperature: 40
1.117	BSS138W-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2601	50V-0.34A-SOT323	1	0.004199	0.004199	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.15, Percentage: 6.666666667	Rated Power: 0.15, Percentage: 6.666666667	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.31600413, λG : 3.44, σG : 0.75, λSS : 3.2611626, σSS : 0.71100929, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.118	-	Optical Device	Other	Single LED / LCD Segment	IR1,IR2,IR3,IR4,I R5,IR6,IR7,IR8,I R9,IR10,IR11,IR12,IR13,IR14,IR15,IR16,IR17,IR18	Night Mode	18	0.000917	0.016507	Type: Single LED / LCD Segment, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, Δ G: 0.25, σ G: 0.18, Δ SS: 0.75, σ SS: 0.54, Temperature: 40
1.119	-	Optical Device	Isolators, Dual / Couplers	Light Sensitive Resistor	LDR1	Ambient Light Sensor, Light Dependent Resistor	1	0.123111	0.123111	Type: Light Sensitive Resistor, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, Δ G: 21, σ G: 15, Δ SS: 63, σ SS: 45, Temperature: 40
1.120	[HFD42/4.5-3SR(825)]-DC5V-HF	Relay	General Purpose		K1	HFD42/4.5-3SR(825)	1	0.009483	0.009483	Quality Level: Level I, Contact Current: 1, Rated Current: 4, Percentage: 25	Contact Current: 1, Rated Current: 4, Percentage: 25	Level I	π Q: 3, π E: 1, π T: 1, Δ S: 0.72252735, Δ G: 1.9, σ G: 1.9, Δ SS: 4.1184059, σ SS: 4.1184059, Temperature: 40
1.121	TSTP650K-180BH-8.4	Switch	Pushbutton		SW2	DC12V-50mA-4PIN-6*6*5mm	1	0.007842	0.007842	Quality Level: Level I, Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Level I	π Q: 3, π E: 1, π T: 1, Δ S: 0.67705687, Δ G: 1.99, σ G: 1.4, Δ SS: 4.0420295, σ SS: 2.8436389, Temperature: 40
1.122	K-F-24.000M-9-2020-4085-80	Miscellaneous	Crystal Oscillator		X803	24MHz $\pm$ 20ppm-(-40°C-85°C)-3.2*2.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, Δ G: 3.2, σ G: 1.5, Δ SS: 9.6, σ SS: 4.5, Temperature: 40
1.123	SF32WK32768D-31T002	Miscellaneous	Crystal Oscillator		X802	32.768K $\pm$ 20ppm-(-40°C-85°C)-3.2*1.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, Δ G: 3.2, σ G: 1.5, Δ SS: 9.6, σ SS: 4.5, Temperature: 40
1.124	YT30028FB-IR-JZ-DK-	Rotating Device				Lens-Electric Zoom	2	1.000000	2.000000				
1.125	IR0109	Rotating Device				Day and Night Mode	1	1.000000	1.000000				
1.126	ETA505252G	Integrated Circuit	Analog		U2	[ETA505252G/E TA5052V052G]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, Δ G: 5.3630921, σ G: 3.7917061, Δ SS: 16.089276, σ SS: 11.375118, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.127	LP3981AB5F]-SOT23-5	Integrated Circuit	Analog		U3,U7	[LP3981AB5F]-SOT23-5	2	0.026871	0.053741	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.128	[TP10-2-VR]-VSSOP8	Integrated Circuit	Analog		U4	[TP10-2-VR]-VSSOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.129	[TPM16050-S6TR/TM18113]-SOT23-6	Integrated Circuit	Analog		U5	[TPM16050-S6TR/TM18113]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.130	SGM431XN3LG	Integrated Circuit	Analog		U13,U15	[LT1431APTSLT1G]-SOT23	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.131	TL2845BDR-8	Integrated Circuit	Analog		U12	[TL2845BDR-8/SA2845SATR/SAP2845SATR]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.132	DIO54335TST6]-SOT23-6-dioo	Integrated Circuit	Analog		U709	[DIO54335TST6]-SOT23-6	1	0.028473	0.028473	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40
1.133	ETA3485S2F]-SOT23-5-ETA	Integrated Circuit	Analog		U702,U703,U706	[ETA3485S2F]-SOT23-5	3	0.022631	0.067893	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40

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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.134	ETA5053V180D F1E]-1.8V- DFN1*1-ETA	Integrated Circuit	Analog		U710	[ETA5053V180D F1E]-1.8V- DFN1*1	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.135	LP3981A85F]- SOT23-5	Integrated Circuit	Analog		U707	[LP3981A85F]- SOT23-5	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.136	TM15302]- TSOT23-6	Integrated Circuit	Analog		U6	[TM15302]- TSOT23-6	1	0.035814	0.035814	Quality Level: Level I, Number of Transistors: 768		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.1386501, σ G: 4.3400256, λ SS: 18.41595, σ SS: 13.020077, Temperature: 40
1.137	SD49245TR]- S08	Integrated Circuit	Analog		U10	[SD49245TR]- SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.138	EL357N(B)(TA)- VG	Integrated Circuit	Analog		U11	[EL357N(B)(TA)(BMS)-G]-SO6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.139	TP8485E-SR	Integrated Circuit	Analog		U20	[TP8485E- SR/SIT3088EESA]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.140	MS41908M	Integrated Circuit	Analog		U36	[MS41908M]- QFN44	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.141	IMX307LQR-C	Integrated Circuit	Digital		U1	[IMX307]LQR-C]- 2.0MP-1/2.8- LGA110	1	0.031007	0.031007	Quality Level: Level I, Technology: CMOS, Number of Gates: 5000000		Level I	πQ: 3, πE: 1, πT: 1, λG: 6.9374105, σG: 2.5470825, λSS: 20.812232, σSS: 7.6412476, Temperature: 40
1.142	W25N01GVZEIG	Integrated Circuit	ROM, PROM, EPROM		U1101	[W25N01]GVZEI G]- WSON8 (8x6mm)-3V3	1	0.249449	0.249449	Quality Level: Level I, Technology: CMOS, Number of Bits: 1, Bit Units: Gb		Level I	πQ: 3, πE: 1, πT: 1, λG: 42.756504, σG: 30.228848, λSS: 128.26951, σSS: 90.686545, Temperature: 40
1.143	NT5CB256M16E R-FL	Integrated Circuit	RAM		U1001	[NT5CB256M16 ER-FL]-FBGA96	1	0.115747	0.115747	Quality Level: Level I, Technology: Dynamic NMOS/CMOS, Number of Bits: 4, Bit Units: Gb		Level I	πQ: 3, πE: 1, πT: 1, λG: 20.379802, σG: 13.601017, λSS: 61.139406, σSS: 40.80305, Temperature: 40
1.144	C5-36GW	Integrated Circuit	Microprocessor		U801	[C5-36GW]- BGA307	1	0.154317	0.154317	Quality Level: Level I, Technology: CMOS, Bus Width in Bits: 32		Level I	πQ: 3, πE: 1, πT: 1, λG: 24.604507, σG: 20.224777, λSS: 73.81352, σSS: 60.674332, Temperature: 40
1.145	S16403	Inductor	Transformer	Power (> 1W)	T1	[S16403]100M POE-1 channel	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λG: 5, σG: 2, πQ: 3, πE: 1, πT: 1, λG: 5, σG: 2, λSS: 15, σSS: 6, Temperature: 40
1.146	GEP13	Inductor	Transformer	Power (> 1W)	T2	GEP13-127μH	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λG: 5, σG: 2, πQ: 3, πE: 1, πT: 1, λG: 5, σG: 2, λSS: 15, σSS: 6, Temperature: 40
1.147	BA301N	Miscellaneous	Surge Arrestor SMD / Gas Discharge Tube			Operational Amplifier	1	0.002596	0.002596	Quality Level: Level I		Level I	πQ: 3, πE: 1, λG: 0.49, σG: 0.28, λSS: 1.47, σSS: 0.84
1.148	[ML414H- IV01E]-3V	Miscellaneous	Battery		BAT801	Rechargeable, 1mAh	1	0.104891	0.104891	Battery Type: Lithium, Quality Level: Level I		Level I	πQ: 3, πE: 1, λG: 25, σG: 7.5, λSS: 75, σSS: 22.5
1.149	Fuse-4A	Miscellaneous	Fuse		FU1	[6125B4A250V -GI-4A-250V 6125	1	0.001964	0.001964	Fuse Type: ≤ 30A, Quality Level: Level I		Level I	πQ: 3, πE: 1, λG: 0.4, σG: 0.19, λSS: 1.2, σSS: 0.57



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